

Week 8 Summary Report

Project - E-Commerce Order Analytics System

Objective

Design and develop an end-to-end e-commerce order analytics system combining Python and SQL, from dataset generation to business reporting.

Step 1: Generate Realistic Datasets

A Python script generated customers, products, orders and order_items datasets using random/fake values. Real-world inconsistencies such as NULL customer IDs, invalid email formats, incorrect date formats, duplicate records, orphan order items, negative quantities and invalid discounts were intentionally introduced. The generated datasets were exported as CSV files into the data/raw directory.

Step 2: Data Cleaning with Pandas

Loaded raw CSV files into Pandas DataFrames, handled missing values, removed duplicates, standardized product names, corrected date formats, validated emails, checked referential integrity and exported cleaned CSV files.

Step 3: Load into SQL Database

Created SQLite schema with Primary Keys, Foreign Keys, NOT NULL and CHECK constraints. Loaded cleaned datasets and verified table relationships and row counts.

Step 4: SQL Analytics – Joins & Aggregations

Implemented JOIN and aggregation queries to calculate revenue by customer/category/month, top products and Average Order Value (AOV).

```
=====
                        REVENUE REPORT
                        Period: 2026-06-01 to 2026-06-30
=====
```

[TOTAL REVENUE BY CATEGORY]

Category	Total Revenue
Home	90,194.70
Clothing	80,592.95
Electronics	73,947.60
Books	67,705.09

[REVENUE BY CUSTOMER, CATEGORY, MONTH (Top 15)]

Cust ID	Customer Name	Category	Month	Revenue
CUST00151	Kevin White	Clothing	2026-06	7,604.90
CUST00046	John Lewis	Home	2026-06	6,107.81
CUST00340	Emma Wilson	Clothing	2026-06	5,727.56
CUST00375	Michael Brown	Clothing	2026-06	5,452.16
CUST00187	George Smith	Clothing	2026-06	5,428.18
CUST00094	Diana Harris	Books	2026-06	5,416.31
CUST00483	Diana Hall	Home	2026-06	5,301.80
CUST00152	Sarah Clark	Clothing	2026-06	4,789.90
CUST00375	Michael Brown	Electronics	2026-06	4,736.93
CUST00200	Laura Brown	Electronics	2026-06	4,692.27
CUST00337	Fiona White	Clothing	2026-06	4,649.43
CUST00119	Emma Lee	Electronics	2026-06	4,614.12
CUST00335	Diana Lewis	Home	2026-06	4,491.55
CUST00370	Bob Smith	Books	2026-06	4,139.07
CUST00551	Ethan Miller	Books	2026-06	4,128.05

Step 5: Window Functions & CTEs

Used DENSE_RANK(), LAG(), NTILE(), SUM() OVER(), AVG(), OVER() and CTEs for customer ranking, running totals, moving averages and lifetime value analysis.

=====				
TOP CUSTOMERS & RANKINGS REPORT				
Period: 2026-06-01 to 2026-06-30				
=====				
[TOP 10 CUSTOMERS BY SPENDING]				
-----+-----+-----+-----+-----				
Cust ID	Customer Name	Type	Total Spend	Orders
-----+-----+-----+-----+-----				
CUST00151	Kevin White	VIP	15,209.39	3
CUST00375	Michael Brown	PREMIUM	10,995.36	1
CUST00483	Diana Hall	REGULAR	10,739.20	2
CUST00038	Sarah Harris	REGULAR	9,817.13	2
CUST00046	John Lewis	PREMIUM	9,270.20	2
CUST00094	Diana Harris	PREMIUM	8,674.95	1
CUST00367	Alice Johnson	PREMIUM	7,955.34	2
CUST00200	Laura Brown	PREMIUM	7,495.69	1
CUST00476	Diana Smith	REGULAR	7,432.71	2
CUST00410	Julia Robinson	REGULAR	7,340.16	1
-----+-----+-----+-----+-----				
[CUSTOMER DENSE RANK BY LIFETIME VALUE (Top 10)]				
-----+-----+-----+-----+-----				
Cust ID	Customer Name	LTV	Rank	
-----+-----+-----+-----+-----				
CUST00450	Laura Hall	22,764.31	1	
CUST00538	Emma Miller	22,250.14	2	
CUST00193	Julia Garcia	17,239.47	3	
CUST00102	Laura Lewis	16,926.08	4	
CUST00488	John Clark	16,039.51	5	
CUST00384	Bob Davis	15,823.99	6	
CUST00436	John Clark	15,717.71	7	
CUST00153	Ethan Robinson	15,281.24	8	
CUST00151	Kevin White	15,209.39	9	
CUST00012	Fiona Doe	15,123.78	10	
-----+-----+-----+-----+-----				

Step 6: Cohort & Retention Analysis

Created registration and first purchase cohorts, calculated monthly retention rates and identified repeat and churned customers.

=====								
COHORT RETENTION REPORT								
=====								
[COHORT RETENTION BY REGISTRATION MONTH (Top 10 Cohorts)]								
-----+-----+-----+-----+-----+-----+-----+-----+-----								
Cohort	Size	M0	M1	M2	M3	M1 Ret %	M2 Ret %	M3 Ret %
-----+-----+-----+-----+-----+-----+-----+-----+-----								
2025-01	34	4	1	4	1	2.9%	11.8%	2.9%
2025-02	36	0	3	3	1	8.3%	8.3%	2.8%
2025-03	40	1	6	3	5	15.0%	7.5%	12.5%
2025-04	41	2	4	5	2	9.8%	12.2%	4.9%
2025-05	46	3	2	6	4	4.3%	13.0%	8.7%
2025-06	31	2	3	3	1	9.7%	9.7%	3.2%
2025-07	40	3	2	7	3	5.0%	17.5%	7.5%
2025-08	40	3	3	2	3	7.5%	5.0%	7.5%
2025-09	37	2	5	6	8	13.5%	16.2%	21.6%
2025-10	38	2	6	5	3	15.8%	13.2%	7.9%
-----+-----+-----+-----+-----+-----+-----+-----+-----								
[COHORT RETENTION BY FIRST PURCHASE MONTH (Top 10 Cohorts)]								
-----+-----+-----+-----+-----+-----+-----+-----+-----								
Cohort	Size	M0	M1	M2	M3	M1 Ret %	M2 Ret %	M3 Ret %
-----+-----+-----+-----+-----+-----+-----+-----+-----								
2025-01	4	4	0	0	0	0.0%	0.0%	0.0%
2025-02	1	1	0	0	0	0.0%	0.0%	0.0%
2025-03	8	8	0	0	0	0.0%	0.0%	0.0%
2025-04	12	12	3	0	1	25.0%	0.0%	8.3%
2025-05	9	9	0	1	1	0.0%	11.1%	11.1%
2025-06	17	17	2	1	0	11.8%	5.9%	0.0%
2025-07	15	15	1	1	2	6.7%	6.7%	13.3%
2025-08	18	18	4	3	0	22.2%	16.7%	0.0%
2025-09	20	20	1	2	3	5.0%	10.0%	15.0%
2025-10	17	17	5	4	4	29.4%	23.5%	23.5%
-----+-----+-----+-----+-----+-----+-----+-----+-----								

Step 7: Customer Segmentation
Segmented customers using purchase frequency, spending tiers and RFM analysis.

```
=====
CUSTOMER SEGMENTATION & RFM SUMMARY REPORT
=====

[AOV BY CUSTOMER SEGMENT (PURCHASE FREQUENCY)]

-----+-----+-----+-----
Segment      | Total Orders | Total Rev  | AOV
-----+-----+-----+-----
One-time     | 211          | 656,105.45 | 3,109.50
Loyal        | 17           | 51,803.17  | 3,047.25
Occasional   | 529          | 1,534,779.97 | 2,901.29
-----+-----+-----+-----

[RFM CUSTOMER SEGMENTS SUMMARY]

-----+-----+-----+-----+-----
RFM Segment   | Cust Count | Avg Recency | Avg Freq | Avg Spend
-----+-----+-----+-----+-----
Lost Customer | 137        | 202.50      | 1.00     | 3,052.43
Active Engaged | 113       | 39.70       | 2.50     | 6,516.32
Regular       | 76         | 40.60       | 1.00     | 3,138.99
At Risk Loyal | 76         | 165.90      | 2.50     | 7,532.99
Core Loyal    | 25         | 13.70       | 2.70     | 11,083.62
-----+-----+-----+-----+-----
```

Step 8: CLI Reporting Tool

Built report_cli.py supporting reports like daily, monthly, revenue, retention and top_customers with formatted console tables.

```
=====
E-COMMERCE ORDER ANALYTICS SUMMARY REPORT
Report Type: DAILY
Current Period: 2026-06-01 to 2026-06-05
Previous Period: 2026-05-27 to 2026-05-31
=====

[KEY PERFORMANCE METRICS]
-----+-----+-----+-----
Metric      | Current Period | Previous Period | % Change
-----+-----+-----+-----
Total Orders | 21             | 27             | -22.22%
Net Revenue  | 68,851.22      | 84,994.10     | -18.99%
Unique Customers | 20            | 26            | -23.08%
-----+-----+-----+-----

[TOP 3 PRODUCTS BY REVENUE]
-----+-----+-----+-----
Product Name | Category | Units Sold | Revenue
-----+-----+-----+-----
Sneakers 455 | Clothing | 13         | 6,614.95
Sneakers 888 | Clothing | 8          | 4,789.90
Rug 864      | Home    | 7          | 4,491.55
-----+-----+-----+-----

[PERIODIC BREAKDOWN (DAILY)]
-----+-----+-----+-----
Period      | Orders | Revenue | Unique Customers
-----+-----+-----+-----
2026-06-01 | 6      | 20,515.93 | 6
2026-06-02 | 4      | 14,901.69 | 4
2026-06-03 | 5      | 12,734.94 | 5
2026-06-04 | 3      | 12,593.76 | 3
2026-06-05 | 3      | 8,104.89  | 3
-----+-----+-----+-----
```

Step 9: Edge Case Handling

Validated orphan records, future dates, invalid discounts, zero quantity orders, empty results and invalid CLI inputs.

Step 10: Documentation & Submission

Prepared README explaining architecture, folder structure, setup, execution steps, sample outputs and GitHub repository structure.

Conclusion

The project demonstrates a complete end-to-end analytics pipeline including data generation, cleaning, SQL analytics, customer segmentation, cohort analysis and automated reporting.